

Various notes on MLS Chapter 2, Section 4

On eq.s (2.51) & (2.52) p. 52:

In deriving eq. (2.51), after writing the first sub-equation

$$v_{q_a}(t) = \hat{\omega}_{ab}^s R_{ab}(t) q_b$$

MLS substitute $q_b = R_{ab}^{-1}(t) q_a(t)$ (the inverse of the unnumbered 1st equation in §4.1 (p. 51)) into this, to render the second sub-equation of eq. (2.51), which is perhaps cleaner written as

$$v_{q_a}(t) = \hat{\omega}_{ab}^s q_a(t)$$

Also, eq. (2.52) is written in a cleaner way as

$$v_{q_b}(t) = R_{ab}^T(t) v_{q_a}(t) = \hat{\omega}_{ab}^b q_b$$

Both these equations express the same vector cross product calculation of the velocity vector of point q :

$$\vec{v}_q = \vec{\omega} \times \vec{q},$$

but in vector component form in two different bases: eq. (2.51) in the base associated with the spatial or inertial frame A and eq. (2.52) in the base associated with the body frame B . (In the above, $\vec{v}_q$ and $\vec{q}$ are the velocity and position vectors of point q , respectively, and $\vec{\omega}$ is the angular velocity vector of the rotating-only rigid body of which q is a part.)

On eq. (2.53) p. 54:

Since p_{ab} are the coordinates of the origin of frame B in the spatial frame, one can interpret the term $\dot{p}_{ab}$ that appears in eq. (2.53) as the velocity of the origin of frame B , in components in the spatial frame.

On an unnumbered equation on p. 54:

The unnumbered equation $q_a(t) = g_{ab} q_b$ following equation (2.53) is essentially the unnumbered equation just preceding equation (2.23) on p. 36.

On eq. (2.54) p. 54:

One must be very careful in reading this equation. Strictly speaking it does not make mathematical sense. The first part of the equation

$$v_{q_a} = \hat{V}_{ab}^s q_a$$

expresses a 4×1 column vector as a 4×4 matrix post-multiplied by a 4×1 column vector. The second part of the equation expresses the sum of a cross product and a vector in $\mathbb{R}^3$ and is cleaner written as

$$\hat{\omega}_{ab}^s q_a + v_{ab}^s.$$

This therefore renders a 3×1 column vector. To make sense of this one should therefore interpret the expression $\hat{\omega}_{ab}^s q_a + v_{ab}^s$ as just filling in the first three rows of the 4×1 v_{q_a} , the 4th row being a zero value as v_{q_a} is a vector in homogeneous form.

Note that $v_{q_a}(t)$ in eq. (2.51) p. 52 is 3×1 whereas v_{q_a} in eq. (2.54) p. 54 is 4×1 , the latter being an homogeneous form. Of course $v_{q_a}(t)$ in eq. (2.51) p. 52 was derived for the case where there is no translation (rotation only), while v_{q_a} in eq. (2.54) p. 54 is used when there is rotation as well as translation.

On eq. (2.56) p. 55:

The same can be said of eq. (2.56) p. 55. The first part of the equation

$$v_{qb} = \hat{V}_{ab}^b q_b$$

expresses a 4×1 column vector as a 4×4 matrix post-multiplied by a 4×1 column vector. The second part of the equation expresses the sum of a cross product and a vector in $\mathbb{R}^3$ and is cleaner written as

$$\hat{\omega}_{ab}^b q_b + v_{ab}^b .$$

This therefore renders a 3×1 column vector. To make sense of this one should therefore interpret the expression $\hat{\omega}_{ab}^b q_b + v_{ab}^b$ as just filling in the first three rows of the 4×1 v_{qb} , the 4th row being a zero value as v_{qb} is a vector in homogeneous form.

Proof of Lemma 2.13 p. 56

Let

$$g = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad g^{-1} = \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix}$$

see MLS pp. 36 & 37, and $\xi^T = [v^T \quad \omega^T]$. Then

$$\hat{\xi} = \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix} .$$

Therefore,

$$\begin{aligned} g \hat{\xi} g^{-1} &= \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\omega} & v \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^T & -R^T p \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{\omega} R^T & -\hat{\omega} R^T p + v \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} R \hat{\omega} R^T & -R \hat{\omega} R^T p + R v \\ 0 & 0 \end{bmatrix} \end{aligned} \quad (1)$$

On the other hand,

$$\text{Ad}_g \xi = \begin{bmatrix} R & \hat{p}R \\ 0 & R \end{bmatrix} \begin{bmatrix} v \\ \omega \end{bmatrix} = \begin{bmatrix} Rv + \hat{p}R\omega \\ R\omega \end{bmatrix} . \quad (2)$$

Now, MLS eq 2.50 p. 52 implies

$$\omega^s = R\omega^b \quad \Longleftrightarrow \quad \hat{\omega}^s = R\hat{\omega}^b R^T$$

and $\hat{p}R\omega = -(R\omega)^\wedge p$. Using this when packing the right hand side of equation (2) above, $\in \mathbb{R}^6$, into the 4×4 matrix $\in se(3)$, the matrix on the right hand side of equation (1) above results, which proves the lemma.

Supplementary to Lemma 2.13 p. 56

Consider a special case of Lemma 2.13, where $g = e^{\hat{\xi}\theta}$. For this special case

$$g \hat{\xi} g^{-1} = e^{\hat{\xi}\theta} \hat{\xi} e^{-\hat{\xi}\theta} .$$

But using the Taylor-series expansion of $e^{\hat{\xi}\theta}$ given on p. 40, it is easy to show that

$$e^{\hat{\xi}\theta} \hat{\xi} = \hat{\xi} e^{\hat{\xi}\theta} .$$

Therefore, for this special case,

$$g \hat{\xi} g^{-1} = \hat{\xi} e^{\hat{\xi}\theta} e^{-\hat{\xi}\theta} = \hat{\xi} ,$$

which then also means that for this special case

$$\text{Ad}_g \xi = \xi .$$

On the 1st equation p. 58:

This is eq. (2.45) p. 49.

1st time derivative of matrix exponential, p. 58:

For constant $\hat{\xi}$:

$$\begin{aligned}
 e^{\hat{\xi}\theta} &= I + \hat{\xi}\theta + \frac{(\hat{\xi}\theta)^2}{2!} + \frac{(\hat{\xi}\theta)^3}{3!} + \dots \\
 \frac{de^{\hat{\xi}\theta}}{dt} &= \hat{\xi}\dot{\theta} + \hat{\xi}^2\theta\dot{\theta} + \frac{\hat{\xi}^3\theta^2\dot{\theta}}{2!} + \frac{\hat{\xi}^4\theta^3\dot{\theta}}{3!} + \dots \\
 &= \hat{\xi}\dot{\theta} \left[I + \hat{\xi}\theta + \frac{(\hat{\xi}\theta)^2}{2!} + \frac{(\hat{\xi}\theta)^3}{3!} + \dots \right] = \left[I + \hat{\xi}\theta + \frac{(\hat{\xi}\theta)^2}{2!} + \frac{(\hat{\xi}\theta)^3}{3!} + \dots \right] \hat{\xi}\dot{\theta} \\
 &= \hat{\xi}\dot{\theta}e^{\hat{\xi}\theta} = e^{\hat{\xi}\theta}\hat{\xi}\dot{\theta}
 \end{aligned}$$

Two applications of Lemma 2.13 p. 56:

MLS use Lemma 2.13 in deriving two equations, without stating so:

- The last step in deriving equation $\hat{V}_{ab}^b = \left(\text{Ad}_{g_{ab}^{-1}(0)} \hat{\xi} \right)^\wedge \dot{\theta}$, p. 58.
- The last equation before proposition 2.15, p. 59.